

MUHAMMAD ISMAIL

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SUMMARY

Cybersecurity undergraduate at GIK Institute (BS Cybersecurity, Expected 2027) seeking Summer 2026 internships in security engineering, DevSecOps, SOC operations, or GRC. Solid foundations in information security and assurance (CIA, AAA, risk management), with hands-on experience across penetration testing, vulnerability assessment, threat modeling (STRIDE), CI/CD security automation (SAST, DAST, SCA), Cisco network design, and machine learning for security. Leading SOC Central FYP with PITB as industry stakeholder; completed a Summer 2025 cybersecurity internship at PITB.

EDUCATION

GIK Institute of Engineering Sciences and Technology

Topi, KPK

BS Cybersecurity | Sep 2023 - Jun 2027 (Expected)

Relevant coursework: Information Security & Assurance, Secure Software Design & Engineering, Wireless & Mobile Security, Cryptography, Computer Communication Networks, Design & Analysis of Algorithms, Database Systems, Operating Systems.

EXPERIENCE

Cybersecurity Intern — Punjab Information Technology Board (PITB)

Jul 2025 - Aug 2025

Lahore, Pakistan

- Investigated security alerts and triaged logs alongside SOC analysts in a live Security Operations Center environment, covering incident response procedures and threat detection workflows.
- Gained hands-on experience with enterprise SIEM tooling, vulnerability management, and Linux-based security infrastructure deployed at government scale.

LEADERSHIP & ACHIEVEMENTS

Sponsorship & Liaison Head — GIK Science Society

2024 - 2025

- Negotiated corporate sponsorships and managed stakeholder relationships end-to-end for two successful All-Pakistan-scale technical events, coordinating cross-functional teams (events, marketing, logistics) to deliver on schedule and within budget.

Vice Captain — GIK Tennis Team

2024 - Present

- Co-lead team management, training coordination, and inter-university tournament representation; mentor junior players and support match strategy and roster decisions.

SELECTED PROJECTS

SOC Central — AI-Driven Security Operations Platform

Final Year Project, in progress

Supervisor: Dr. Zubair Ahmad | Industry stakeholder: PITB

- Architecting a centralized SOC consolidating eBPF / Osquery / YARA endpoint detection and response (EDR) telemetry into an XGBoost + SHAP detection engine; microservices on Kubernetes (K3s), enriched with EPSS / CISA KEV threat intelligence, on a 16-week phased roadmap.

AI-Integrated Intrusion Detection System

Machine Learning Security

Python, CatBoost, scikit-learn, SMOTE / ADASYN, Streamlit | github.com/aliuzair1/AI-Integrated-Intrusion-Detection-System | Collaborator

- Built a hybrid AI-based IDS combining CatBoost gradient-boosting with Local Outlier Factor (LOF) unsupervised anomaly detection in an OR-voting ensemble to maximize threat detection recall; applied SMOTE + ADASYN resampling for class imbalance, leakage-aware temporal split, and deployed a Streamlit inference dashboard with serialized model artifacts.

AUI Secure — DevSecOps CI/CD Pipeline & Secure Coding Platform

CY-321

Python (Flask 3.0), GitHub Actions, SQLite, Bandit, OWASP ZAP, Trivy

- Built a Flask app demonstrating OWASP Top 10 with paired vulnerable / secure endpoints, plus a 5-stage GitHub Actions pipeline (tests, secrets, SAST, SCA, DAST) with PR blocking; authored STRIDE Threat Model and Risk Management deliverables.

Secure IoT / Edge Network with Zero-Trust Architecture

CE313 — CCN

Cisco Packet Tracer, Cisco IOS, Linux iPerf3 testbed | Team project

- Designed a three-tier enterprise network for a 15,000 m² smart factory with 50+ IoT endpoints under NIST SP 800-207 Zero-Trust; 8-VLAN, OSPF, 802.1X / RADIUS, DAI, ASA / pfSense; HTTP/3 delivered 92% throughput gain over HTTP/2 at 5% loss.

Sportify (TurfBook) — Co-Founder & Builder, Football Turf Booking & Matchmaking

Entrepreneurship Course

React 18 + TypeScript, Express.js, PostgreSQL (Neon), Drizzle ORM, Tailwind, shadcn/ui | github.com/kryptbakar/sportify

- Co-built a full-stack TypeScript web platform with interval-scheduling booking conflict detection, weighted ELO matchmaking, four-tier ranking (K-factor 32), OpenID Connect RBAC, and Zod-validated REST API; deployed on Railway.

Also built: DQN-Enhanced AN Power Allocation — Deep Reinforcement Learning agent (TensorFlow 2) for MISO wiretap channels under imperfect CSI (github.com/MuhammadIsmail009/pls-dqn-miso) | Cybersecurity Log Analyzer (C++ — linked lists, hash tables, BSTs, priority queues) | Secure Channel Communication (64-bit encryption, WhatsApp API key delivery) | Wi-Fi Disruption Tool (ESP8266 deauth-frame penetration testing) | CEH System Hacking Lab (privilege escalation, password attacks, post-exploitation, Kali Linux)

SKILLS

Languages & Frameworks: Python, C++, TypeScript, JavaScript, SQL (PostgreSQL, SQLite), HTML / CSS, LaTeX, Bash, PowerShell, Flask, Express.js, React 18, Drizzle ORM, Node.js

Security & DevSecOps: Penetration Testing, Vulnerability Assessment, Network Security, STRIDE Threat Modeling, Threat Intelligence, OWASP Top 10, NIST SP 800-207 Zero Trust, ISO 27001, Information Assurance, Risk Management, SOC operations, Incident Response, SIEM, Cryptography, SAST / DAST / SCA | Nmap, Burp Suite, Metasploit, Wireshark, John the Ripper, OWASP ZAP, Bandit, Trivy, CodeQL, Postman, Docker, Kubernetes (K3s), Kali Linux

Networking & AI/ML: Cisco Packet Tracer, Cisco IOS, OSPF / BGP / EIGRP, VLANs, 802.1X / RADIUS, WireGuard / IPsec, WPA2 / WPA3, MQTT | Artificial Intelligence, Deep Learning, Reinforcement Learning, TensorFlow 2 / Keras, scikit-learn, XGBoost, SHAP, Deep Q-Networks, NumPy, Pandas

References: Available upon request.